4.4 Writing Equations of Parallel and Perpendicular Lines

Name _____ Period ____ Date _____

Algebra I 4.4 (30-minute plan) · Student sheet

I can…

- identify parallel and perpendicular lines from their slopes.
- write equations of parallel lines.
- write equations of perpendicular lines.

Vocabulary · fill in the missing word

Parallel lines — lines in the same _____ that never meet; nonvertical parallel lines have equal slopes

Perpendicular lines — lines that meet at right _____; their slopes are negative reciprocals (product −1)

Negative reciprocal — flip the _____ and change the sign: $\frac{2}{3} \rightarrow -\frac{3}{2}$

Part 1 · Notes and examples (fill in each note, then work the example)

Parallel lines: Parallel lines have _____ slopes.

Perpendicular lines: Perpendicular slopes are _____ _____; their product is _____.

Example 1

Which lines are parallel? Which are perpendicular?

a: $y = 3x - 2$ b: $y = -\frac{1}{3}x + 5$ c: $6x - 2y = 8$

Example: Perpendicular to slope ¾: _____.

Example 2

Write an equation of the line through $(-4, 1)$ perpendicular to $y = 2x - 7$.

Part 2 · Partner practice

Partner practice

Write the line through $(0, -2)$ parallel to $2x + y = 6$.